Date:23.11.2020

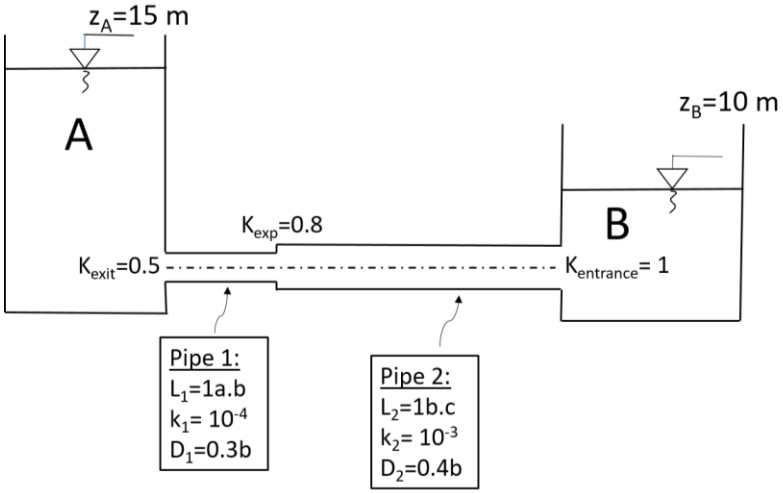
2020-2021 FALL SEMESTER HYDRAULICS COURSE

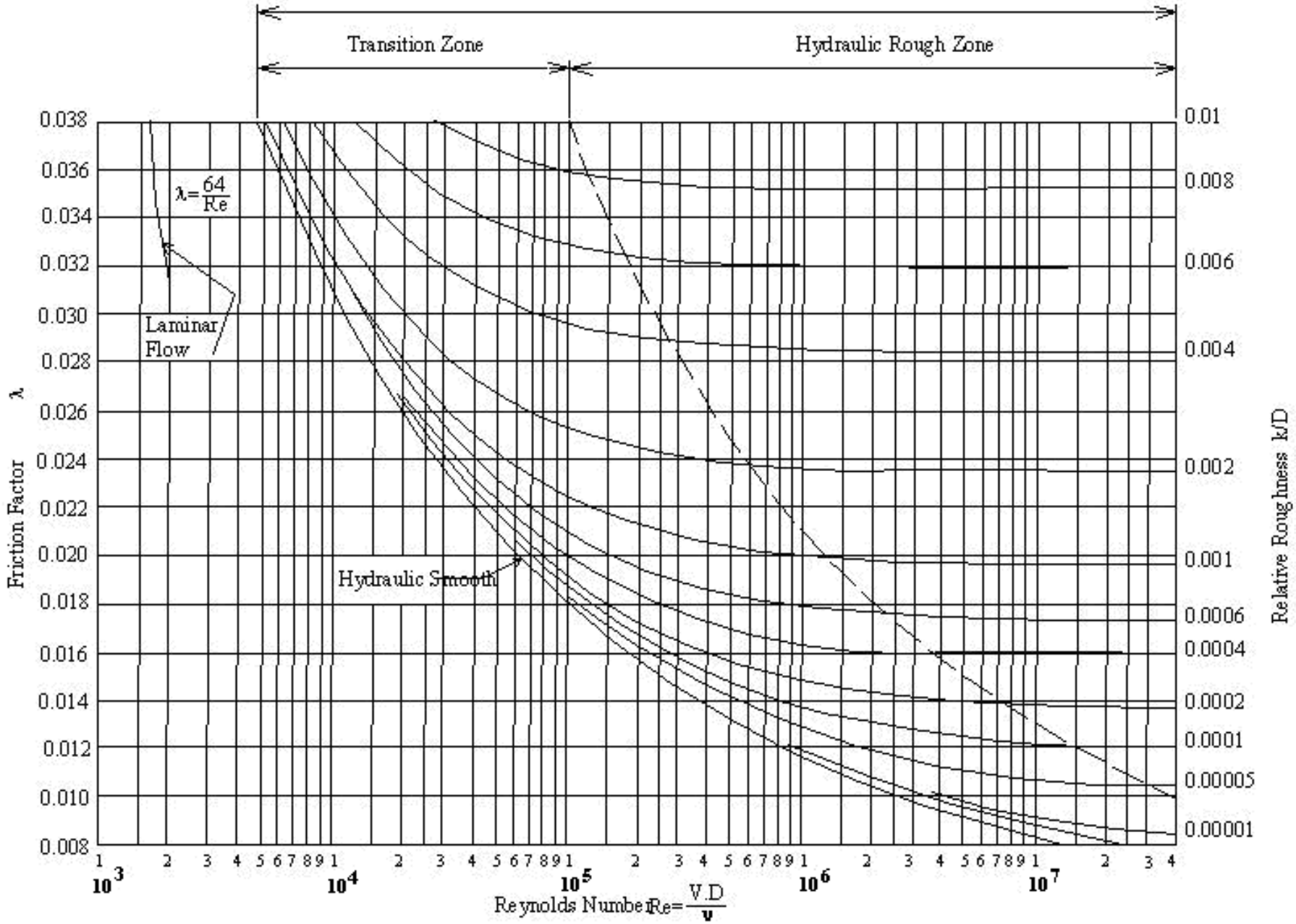
Name: **Surname:** **Student ID:**

Question 2 (25p):

Calculate the velocities and the flowrates of each pipe (Pipe 1 and Pipe 2) shown in the figure. Draw both the energy grade line (EGL) and hydraulic grade line (HGL) on the figure. The kinematic viscosity of the fluid is $\boldsymbol{\nu = 10^{-7}}\ m^2/s$. (g=10 m/s^2)

Important Note: All the lengths in the figure are in **meter**s. L, k, and D are the length of pipe, the roughness height, and the diameter of the pipes, respectively. Where c,b, and a are the last three digits of your student ID. For example, if the student ID is 1234**567**, **c is 5**, **b is 6** and **a is 7**. For this student ID, for example, the length are as follows: L_1=17.6 m, L_2= 16.5 m, D_1= 0.36 m, and D_2=0.46 m. The roughness heights in the pipes are constant and equal to $k_1=10^{-4}$ m in Pipe1 and $k_2=10^{-3}$ m in Pipe 2.




Transition Zone
Hydraulic Rough Zone
λ=64/Re
Laminar Flow
Hydraulic Smooth
Friction Factor λ
Relative Roughness k/D
Reynolds Number Re=V.D/ν
0.038
0.036
0.034
0.032
0.030
0.028
0.026
0.024
0.022
0.020
0.018
0.016
0.014
0.012
0.010
0.008
0.01
0.008
0.006
0.004
0.002
0.001
0.0006
0.0004
0.0002
0.0001
0.00005
0.00001
10^3
10^4
10^5
10^6
10^7